



CT-based deep learning model for predicting the success of extracorporeal shock wave lithotripsy in treating ureteral stones larger than 1 cm

Rijin Song¹ · Bo Liu² · Huixin Xu¹

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Abstract

Objectives To develop a deep learning (DL) model based on computed tomography (CT) images to predict the success of extracorporeal shock wave lithotripsy (SWL) treatment for patients with ureteral stones larger than 1 cm.

Materials and methods We enrolled 333 patients who underwent SWL treatment for ureteral stones and randomly divided them into training and test sets. A DL model was built based on CT images of ureteral stones to predict SWL outcomes. The predictive efficacy of the DL model was assessed by comparing it with traditional and radiomics models.

Results The DL model demonstrated significantly better predictive performance in both training and test sets compared to radiomics (training set, AUC: 0.993 vs. 0.923, $P < 0.001$; test set AUC: 0.982 vs. 0.846, $P < 0.001$) and traditional models (training set AUC: 0.993 vs. 0.75, $P = 0.005$; test set AUC: 0.982 vs. 0.677, $P < 0.001$). Decision curve analysis (DCA) also proved that the DL model brought more benefit in predicting the success of SWL treatment than other methods.

Conclusion The DL model based on CT images showed excellent ability to predict the probability of success of SWL treatment for patients with ureteral stones larger than 1 cm, providing a new auxiliary tool for clinical treatment decision-making.

Keywords Deep learning · Shock wave lithotripsy · Ureteral stones · Treatment outcome · Prediction

Introduction

Extracorporeal shock wave lithotripsy (SWL) and ureteroscopy (URS) are the leading methods for treating ureteral stones. Compared to URS, SWL is associated with fewer overall complications but a higher rate of retreatment and the need for secondary procedures [1]. Therefore, many

efforts have been made to reduce SWL failure rates by identifying appropriate SWL outcome predictors, and non-contrast computed tomography (CT) has played a crucial role in these attempts [2]. The European Association of Urology (EAU) guidelines currently endorse URS as the first line of treatment for patients with ureteral stones larger than 1 cm [1]. However, SWL may also be recommended for appropriately selected patients in clinical practice.

A number of studies have shown that certain CT features of urinary stones are correlated with their fragility and can help predict outcomes of SWL treatment [3–7]. However, the predictive power of some of the relevant CT features reported by these studies remains controversial, such as CT attenuation value and texture characteristics [8, 9]. Radiomics allows the extraction of a large number of quantitative imaging features from medical images, which are difficult to discern with the naked eye [10]. In addition to texture characteristics, radiomics can provide morphological features that can be used as an indicator of urinary stone fragility in SWL [11]. Despite limited literature exploring

✉ Huixin Xu
xuhuixin1123@126.com

Rijin Song
songrijin@163.com

Bo Liu
lb_xhx@126.com

¹ Department of Urology, The First Affiliated Hospital of Nanjing Medical University, No. 300 Guangzhou Road, Nanjing, Jiangsu Province, China

² Department of Radiology, The First Affiliated Hospital of Nanjing Medical University, No. 300 Guangzhou Road, Nanjing, Jiangsu Province, China

the application of radiomics to urolithiasis, existing research results have shown promising application prospects [12].

Recently, deep convolutional neural networks (CNNs), a type of deep learning (DL) technique that allows an algorithm to learn the appropriate feature based on medical imaging datasets, have been extensively discussed and have achieved significant results [13, 14]. However, at present, only a few studies have applied DL methods in the field of urolithiasis [15, 16]. Our hypothesis is that DL may be able to extract more valuable information from CT images of stones, thus providing a more accurate estimate of an individual's likelihood of stone clearance and subsequently guiding accurate treatment strategy decisions.

Therefore, we aimed to establish and validate a model based on DL features extracted from CT images to predict the probability of success of SWL treatment for ureteral stones larger than 1 cm, and to compare its predictive performance with traditional and radiomics models.

Patients and methods

This retrospective study was approved by the Institutional Review Board of all participating hospitals, and the requirement for informed consent was waived.

Patients and data collection

Clinical and imaging data from patients who received SWL treatment in our hospital between January 2018 and December 2023 were collected. Inclusion criteria included: (a) with non-contrast CT images prior to treatment; (b) with complete follow-up data; (c) solitary ureteral stones with a maximum diameter larger than 1 cm. A total of 333 patients met the criteria and were included in the study. Patients were randomly divided into training ($n=266$) and test sets ($n=67$) with a ratio of 4:1.

Clinical data collected included age, gender, body mass index (BMI), stone location, laterality, diabetes mellitus (DM), hypertension (HTN), hydronephrosis grade (HG), and skin-to-stone distance (SSD). SWL was performed as previously described using the Dornier Compact Delta II lithotripter (Dornier MedTech, Wessling, Germany) by the same experienced urologist. Patients were administered with appropriate energy levels of shock waves at a low frequency of 60–90 shock waves per minute, with stepwise power ramping, depending on the stone location, treatment response and patient tolerance, until the target stone was completely fragmented or safe energy limits were reached. Treatment outcomes were assessed two weeks after each SWL session using kidney, ureter, and bladder (KUB) film and ultrasound. If after three SWL sessions, the stone failed

to disintegrate or disintegrate sufficiently, and a subsequent intervention (SWL or URS) was required, the patient was considered a treatment failure. Treatment success was defined as complete absence of stones.

Traditional model

According to the previous research results, the predictors of stone CT characteristics used in the construction of traditional models in this study include maximum craniocaudal diameter (MCD), maximum transverse diameter (MATD), minimum transverse diameter (MITD), stone volume (SV), and mean stone density (MSD), which are widely used in clinical practice to predict SWL treatment outcomes.

Radiomics feature extraction

Three-dimensional stone segmentation was semi-automatically performed using a threshold segmentation method using ITK-SNAP software (version 3.8.0, <http://www.itksna.org/>). Before feature extraction, images were resampled to a pixel of $1 \times 1 \times 1$ mm. The radiomics features were extracted using PyRadiomics package (<https://pyradiomics.readthedocs.io>; version 3.0.1) embedded in Python.

The radiomics features included: (I) geometry, (II) intensity, and (III) texture. Geometrical features describe the three-dimensional shape characteristics of the stone. Intensity characteristics describe the first-order statistical distribution of voxel intensities within the stone. Textural features describe patterns, or spatial distributions of intensities in the second and higher order. The texture features are extracted using several different methods, including gray-scale cooccurrence matrix (GLCM), grayscale run length matrix (GLRLM), grayscale size zone matrix (GLSZM), and neighborhood grayscale tone difference matrix (NGTDM) methods.

Deep learning feature extraction

The CNN model, which was pre-trained on ImageNet dataset and retrained on each training dataset via transfer learning, was used for DL features extraction. The base model, which functions as a feature encoder, greatly influenced classification efficiency. To determine the most suitable DL framework for our prediction task, we compared the performance of DenseNet121, Inception V3, ResNet50, ResNet101 and VGG 19. ROI images fed to CNNs were cropped from axial planes with maximum dimensions, which were extended by a buffer of 10 voxels based on the stone segmentation mask. After the optimal base model selected, the ROIs cropped from coronal planes with maximum dimensions were further used as the input. We compared the performance of the

two CNN models based on the two different image inputs and then used the superior model as the feature extractor.

The training set was used to optimize the model parameters. We used the cross-entropy loss function and adopted SGD optimizer for training, with a learning rate of 0.01 for 50 epochs using a batch size of 32. Finally, DL features were extracted from the penultimate layer of the fully connected (FC) layer of the fine-tuned CNN.

Construction of radiomics and DL models

First, Z-score normalization was applied to standardize the extracted feature data.

To establish the radiomics model, univariate analysis was performed to identify significant features. Next, redundant features with Spearman's correlation coefficient ≥ 0.8 were eliminated. Finally, the least absolute shrinkage and selection operator (LASSO) regression was used to select predictive features with non-zero coefficients for model construction. To build the DL model, we first used principal component analysis (PCA) to reduce feature dimensionality and then employed LASSO regression to select the most robust features.

Prediction models were established using the support vector machine (SVM) machine learning classifier, which is based on the radial basis function kernel. In the process of fine-tuning the models, a five-fold cross-validation strategy was used to evaluate the performance of the prediction models, and the grid search method was adopted to select the optimal hyperparameters.

Supplementary Fig. 1 shows the image acquisition, processing, and modeling pipeline.

Statistical analysis

All statistical analyses were performed using R software (version 4.1.2; <https://www.r-project.org>) and Python software (version 3.9.7; <http://www.python.org>). Bilateral *p*-values less than 0.05 were considered statistically significant. One-way variance analysis or Mann-Whitney U test was used for continuous variables, and chi-square test or Fisher's exact test was used for categorical variables. Model performance was evaluated using the area under the receiver operating characteristic curve (AUC), accuracy (ACC), sensitivity (SENS), specificity (SPEC), positive predictive value (PPV), negative predictive value (NPV), and F1 score. Statistical differences in predictive SWL outcomes among models were evaluated using Delong's test. Decision curve analysis (DCA) was used to evaluate the clinical utility of the models by quantifying net benefits.

Results

Clinical characteristics

The clinical characteristics of the patients in the training and test sets are summarized in Supplementary Table 1. There was no significant difference in clinical characteristics between the training and test sets [age (years): 47.30 ± 14.81 vs. 46.64 ± 14.54 , $P=0.743$; BMI: 24.58 ± 3.37 vs. 25.08 ± 2.71 , $P=0.262$; SSD (mm): 108.79 ± 13.96 vs. 107.39 ± 11.81 , $P=0.829$; gender (male/female): 209/57 vs. 49/18, $P=0.332$; laterality (right/left): 124/142 vs. 32/35, $P=0.892$; location (proximal/middle/lower): 173/40/53 vs. 50/9/8, $P=0.260$; Diabetes (Yes/No): 29/237 vs. 8/59, $P=0.828$; Hypertension (Yes/No): 44/222 vs. 14/53, $P=0.471$]. In addition, there were no significant differences in clinical characteristics between the successful and unsuccessful groups in the training and testing sets.

Traditional model

After univariate analysis, we identified five significant predictors: MCD, MATD, MITD, SV, and MSD, as detailed in Supplementary Fig. 2. SV had a high Spearman correlation coefficient with other features, so we excluded SV for further model construction. Based on the logistic regression (LR) classifier, we constructed a traditional model using the remaining four predictors (MCD, MATD, MITD, and MSD).

Radiomics and DL models

As shown in Supplementary Table 3, the ResNet50 model performed best in our prediction task when using axial CT images of stones as input compared to other models, with AUC of 0.924 and 0.764 in the training and test sets, respectively. Therefore, we chose ResNet50 as the base model. As shown in Supplementary Table 3, a slight improvement in prediction performance was achieved using coronal CT images as input (axial vs. coronal AUC: training set 0.924 vs. 0.931; test set 0.764 vs. 0.804). Therefore, we used coronal images as input to extract DL features.

Finally, we extracted 1834 radiomics features and 2048 DL features. After reducing feature dimensionality, we selected 19 radiomics features (Supplementary Figs. 2) and 24 DL features fused by PCA (Supplementary Fig. 3), and then used SVM as a classifier to build models, respectively.

Comparison of different models

The predictive performance of traditional, radiomics, and DL models is shown in Table 1. The Delong test showed

Table 1 Performance of different models for predicting SWL success

Cohort	Models	AUC	95% CI	ACC	SENS	SPEC	PPV	NPV	F1 score
Training	Traditional	0.75	0.6886–0.8111	0.741	0.818	0.634	0.754	0.717	0.785
	Radiomics	0.923	0.8927–0.9542	0.85	0.825	0.884	0.907	0.786	0.864
	DL	0.993	0.9862–0.9989	0.962	0.961	0.964	0.974	0.947	0.967
Test	Traditional	0.677	0.5457–0.8077	0.636	0.579	0.714	0.733	0.556	0.647
	Radiomics	0.846	0.7474–0.9443	0.818	0.868	0.75	0.825	0.808	0.846
	DL	0.982	0.9593–1.0000	0.924	0.947	0.893	0.923	0.926	0.935

AUC=area under the receiver operating characteristic curve; CI=confidence interval; ACC=accuracy; SENS=sensitivity; SPEC=specificity; PPV=positive predictive value, NPV=negative predictive value

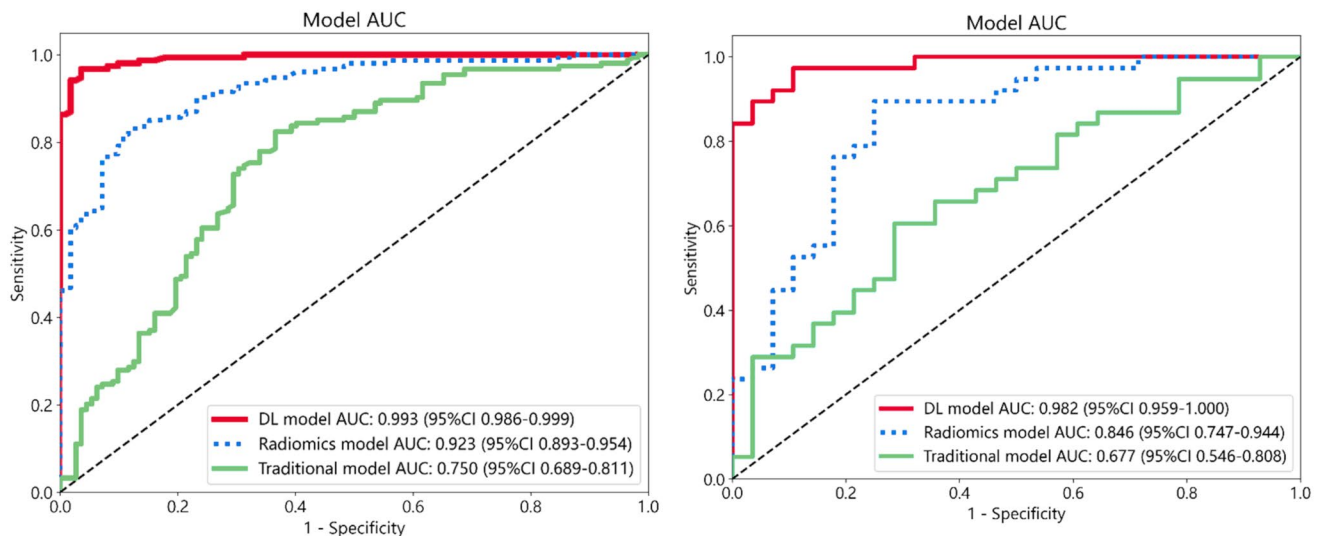


Fig. 1 Receiver operating characteristic (ROC) curves comparison between different models for predicting the success of SWL treatment in the training (a) and test sets (b). DL=deep learning

that the predictive power of the DL model was significantly better than that of the radiomics model (training set AUC: 0.993 vs. 0.923, $P < 0.001$; test set AUC: 0.982 vs. 0.846, $P < 0.001$) and the traditional model (training set AUC: 0.993 vs. 0.75, $P = 0.005$; test set AUC: 0.982 vs. 0.677, $P < 0.001$), and the performance of the radiomics model was also significantly better than that of the traditional model (training set AUC: 0.923 vs. 0.75, $P < 0.001$; test set AUC: 0.846 vs. 0.677, $P = 0.018$). The ROC curves for traditional, radiomics, and DL models in training and test sets are shown in Fig. 1. As shown in Supplementary Fig. 4, in training and test sets, the DL model had the most significant net benefit in predicting the outcome of SWL treatment.

To investigate the interpretability of the DL model, Gradient Weighted Class Activation Mapping (Gradient-CAM) was applied to the final convolutional layer of the last res-block to produce a coarse localization map highlighting important regions in the image for our prediction task, as shown in Supplementary Fig. 5. It was found that, in addition to the stone itself, the ureter surrounding the stone and the area above the stone were both valuable for predicting the outcome of SWL treatment.

Discussion

Accurately predicting the likelihood of success for an individual receiving SWL treatment is crucial for selecting the most appropriate treatment. In this study, we constructed a DL model based on CT images for preoperative prediction of successful SWL treatment of ureteral calculi larger than 1 cm. In addition, the performance of this model outperforms traditional and radiomics models.

In recent years, various DL algorithms have been explored for medical image analysis, where CNNs have become a popular approach for lesion segmentation, classification and prognosis estimation [17–19]. However, their successful implementation requires a large amount of training data, and obtaining a large number of medical images is a challenge in clinical practice [20, 21]. In present study, the number of ureteral stones treated with SWL was limited due to EAU guidelines recommending URS as the primary treatment option for ureteral stones larger than 1 cm [1]. To address this, we used transfer learning to minimize overfitting problems caused by small-scale training data.

After comparing the performance of various predictive models based on different neural network models, including DenseNet121, Inception V3, ResNet101, ResNet50, and VGG 19, we chose to use ResNet50 for our prediction task. ResNet employs residual learning strategies to effectively mitigate accuracy degradation as network depth increases, resulting in improved network accuracy and speed of execution [22]. Although ResNet101 has a deeper network structure, it can capture more complex feature patterns, but in small sample datasets and simple classification tasks like our research, ResNet50 is easier to converge and exhibits better prediction performance, as demonstrated in our study.

3D stone images contain more comprehensive information but require substantial data and higher computing power for effective training in DL analysis [23, 24]. However, given the limited sample size of this study, 3D analysis may not be appropriate. When comparing the performance of two different types of input 2D images, coronal images outperformed axial images in the prediction task in our study, possibly because coronal images provide more comprehensive information about ureteral stones.

This study reveals that the radiomics model also showed good predictive capabilities, but its performance was significantly inferior to that of the DL model. This may be because the CNN training input in this study includes image information around the stone in addition to the stone, whereas the traditional radiomics approach only considers the stone itself. Long-term stone compression in the ureter can lead to ureteral mucosal edema, tissue hypertrophy, polyp formation, stenosis, and even submucosal stone formation, which in turn affects the success rate of SWL treatment [25]. Current studies suggest that insufficient space in the ureter may inhibit shock waves cavitation and reduce the likelihood of stone fragmentation [26]. In addition, stone adhesion to mucosa and lack of expansion space can limit stone discharge. Studies by Sarica K. et al. showed that only 25.2% of patients with impacted proximal ureteral stones became stone free after a single SWL treatment [27]. For predicting ureteral stone impaction, most previous studies have focused on the thickness of the ureter around the stone, and some studies have suggested that the CT value of the ureter wall beneath the stone is an important predictor, as denser ureter tissue may indicate the presence of edema and inflammation [28]. However, it is challenging in clinical settings to measure ureteral thickness manually and accurately, especially in cases where there is a lack of adipose tissue around the mid and distal ureter. Therefore, DL methods that do not require precise segmentation have significant advantages in our prediction tasks.

This study has some limitations: First, as a single-center study, the sample size is limited. The DL model needs to be trained with large-scale sample data to improve accuracy.

In the future, more samples need to be collected and multi-center studies need to be conducted to validate the model. Second, this study is a retrospective study, and there may be bias in sample selection. Further prospective studies are warranted to obtain high-level evidence for clinical use. Third, this study uses two-dimensional features of stones instead of three-dimensional features for prediction. Although two-dimensional feature extraction is simple, it may not fully represent the characteristics of the whole stone, which affects prediction accuracy.

Conclusion

This study constructed and validated a DL model based on CT stone images, which can effectively predict the likelihood of successful SWL treatment of ureteral stones larger than 1 cm, with superior performance comparable to traditional and radiomics models.

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Author contributions R. S. and H. X. collected and wrote the main manuscript text. B. L. and H. X. analysed data. All authors reviewed the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

References

1. Skolarikos A, Neisius HJ, Petřík AA, Tailly T et al (2023) EAU guidelines on urolithiasis
2. Worster A, Preyra I, Weaver B et al (2002) The accuracy of noncontrast helical computed tomography versus intravenous pyelography in the diagnosis of suspected acute urolithiasis: a meta-analysis. *Ann Emerg Med* 40:280–286
3. Guler Y, Erbin A, Kafkasli A et al (2021) Factors affecting success in the treatment of proximal ureteral stones larger than 1 cm with extracorporeal shockwave lithotripsy in adult patients. *Urolithiasis* 49:51–56
4. Langenauer J, Betschart P, Hechelhammer L et al (2018) Advanced non-contrasted computed tomography post-processing by CT-Calculometry (CT-CM) outperforms established predictors for the outcome of shock wave lithotripsy. *World J Urol* 36:2073–2080
5. Niwa N, Matsumoto K, Miyahara M et al (2017) Simple and practical nomograms for predicting the stone-free rate after shock wave lithotripsy in patients with a solitary upper ureteral stone. *World J Urol* 35:1455–1461

6. Yamashita S, Kohjimoto Y, Iguchi T et al (2017) Variation coefficient of Stone Density: a novel predictor of the outcome of extracorporeal shockwave lithotripsy. *J Endourol* 31:384–390
7. Yoshioka T, Ikenoue T, Hashimoto H et al (2020) Development and validation of a prediction model for failed shockwave lithotripsy of upper urinary tract calculi using computed tomography information: the S(3)HoCKwave score. *World J Urol* 38:3267–3273
8. Mannil M, von Spiczak J, Hermanns T et al (2018) Three-dimensional texture analysis with machine learning provides incremental predictive information for successful shock Wave lithotripsy in patients with kidney stones. *J Urol* 200:829–836
9. Xun Y, Li J, Geng Y et al (2018) Single extracorporeal shock-wave lithotripsy for proximal ureter stones: can CT texture analysis technique help predict the therapeutic effect? *Eur J Radiol* 107:84–89
10. Lambin P, Leijenaar RTH, Deist TM et al (2017) Radiomics: the bridge between medical imaging and personalized medicine. *Nat Rev Clin Oncol* 14:749–762
11. Duan X, Qu M, Wang J et al (2013) Differentiation of calcium oxalate monohydrate and calcium oxalate dihydrate stones using quantitative morphological information from micro-computerized and clinical computerized tomography. *J Urol* 189:2350–2356
12. Zheng J, Yu H, Batur J et al (2021) A multicenter study to develop a non-invasive radiomic model to identify urinary infection stone in vivo using machine-learning. *Kidney Int* 100:870–880
13. Tang L, Li W, Zeng X et al (2021) Value of artificial intelligence model based on unenhanced computed tomography of urinary tract for preoperative prediction of calcium oxalate monohydrate stones in vivo. *Ann Transl Med* 9:1129
14. Zeng Q, Feng Z, Zhu Y et al (2022) Deep learning model for diagnosing early gastric cancer using preoperative computed tomography images. *Front Oncol* 12:1065934
15. Huang (2016) Development and validation of a Radiomics Nomogram for Preoperative Prediction of Lymph Node Metastasis in Colorectal Cancer (34, Pg 2157, 2016). *J Clin Oncol* 34:2435–2435
16. Kobayashi M, Ishioka J, Matsuoka Y et al (2021) Computer-aided diagnosis with a convolutional neural network algorithm for automated detection of urinary tract stones on plain X-ray. *BMC Urol* 21:10
17. Kim US, Kwon HS, Yang W et al (2022) Prediction of the composition of urinary stones using deep learning. *Investig Clin Urol* 63:441–447
18. Jiang M, Li C-L, Luo X-M et al (2021) Ultrasound-based deep learning radiomics in the assessment of pathological complete response to neoadjuvant chemotherapy in locally advanced breast cancer. *Eur J Cancer* 147:95–105
19. Kim YI, Song SH, Park J et al (2023) Deep-learning segmentation of urinary stones in Noncontrast Computed Tomography. *J Endourol* 37:595–606
20. Yu J, Deng Y, Liu T et al (2020) Lymph node metastasis prediction of papillary thyroid carcinoma based on transfer learning radiomics. *Nat Commun* 11:4807
21. Huynh BQ, Li H, Giger ML (2016) Digital mammographic tumor classification using transfer learning from deep convolutional neural networks. *J Med Imaging (Bellingham Wash)* 3:034501
22. He K, Ren S, Sun J (2016) Deep residual learning for image recognition. *Proceedings of the IEEE conference on computer vision and pattern recognition*, 770–778
23. Nam JG, Park S, Park CM et al (2022) Histopathologic basis for a chest CT Deep Learning Survival Prediction Model in patients with Lung Adenocarcinoma. *Radiology* 305:441–451
24. Harmon SA, Sanford TH, Xu S et al (2020) Artificial intelligence for the detection of COVID-19 pneumonia on chest CT using multinational datasets. *Nat Commun* 11
25. Mugiya S, Ito T, Maruyama S et al (2004) Endoscopic features of impacted ureteral stones. *J Urol* 171:89–91
26. Pishchalnikov YA, Sapozhnikov OA, Bailey MR et al (2003) Cavitation bubble cluster activity in the breakage of kidney stones by lithotripter shockwaves. *J Endourol* 17:435–446
27. Sarica K, Kafkasli A, Yazici O et al (2015) Ureteral wall thickness at the impacted ureteral stone site: a critical predictor for success rates after SWL. *Urolithiasis* 43:83–88
28. Tran TY, Bamberger JN, Blum KA et al (2019) Predicting the impacted Ureteral Stone with computed tomography. *Urology* 130:43–47

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